



TCP/IP Model

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Introduction to TCP/IP

- The **TCP/IP (Transmission Control Protocol/Internet Protocol)** model is the foundation of modern internet communication. It is a set of networking protocols that allows **devices to communicate over networks, including the Internet**.
- Why is TCP/IP important in Cybersecurity?
 - It defines how **data packets** are transmitted securely.
 - Many **cyberattacks** exploit vulnerabilities in TCP/IP layers.
 - Understanding TCP/IP helps in **network defense, penetration testing, and secure communication**.
- **TCP/IP has four layers:**
 - **1 Application Layer – End-user interaction**
 - **2 Transport Layer – Reliable or fast data transfer**
 - **3 Internet Layer – Addressing and routing**
 - **4 Network Access Layer – Physical transmission of data**

Each layer has specific protocols that facilitate data communication

Layer 1: Network Access Layer (Physical & Data Link Layer in OSI)

- Function:
 - Defines how data is **physically transmitted** over the network.
 - Converts **binary data** into **signals** (electrical, optical, or radio waves).
 - Manages **MAC addresses** for device identification.
- Protocols Used:
 - **Ethernet** – Standard for wired LAN communication.
 - **Wi-Fi (IEEE 802.11)** – Wireless networking protocol.
 - **PPP (Point-to-Point Protocol)** – Used in direct communication links.
 - **ARP (Address Resolution Protocol)** – Resolves **IP addresses to MAC addresses** for network



Layer 2: Internet Layer (Network Layer in OSI)

- Function:
 - Defines **IP addressing and packet routing**.
 - Determines how data **moves across networks**.
 - Uses routers to **forward packets** between networks.
- Protocols Used:
 - **IP (Internet Protocol)** – The primary protocol responsible for addressing.

- **ICMP (Internet Control Message Protocol)** – Used for network diagnostics (**ping, traceroute**).
- **IPsec (Internet Protocol Security)** – Provides **encryption and authentication** for secure communication.
- **RIP (Routing Information Protocol) & OSPF (Open Shortest Path First)** – Used for dynamic routing.
- **Cybersecurity Concerns:**
 - **IP Spoofing:** Attackers modify source IPs to disguise their identity.
 - **ICMP Flood Attack (Ping of Death):** Large ICMP packets cause **Denial of Service (DoS)**.
 - **IP Fragmentation Attacks:** Hackers send fragmented packets to **bypass firewalls**.
- **Security Measures:**
 - **Firewalls & ACLs (Access Control Lists)** – Blocks unauthorized packets.
 - **IPSec VPNs** – Encrypts IP traffic for **secure remote access**.
 - **Disable ICMP on Public Servers** – Prevents ping-based attacks.



Layer 3: Transport Layer (End-to-End Communication)

- **Function:**
 - Ensures **error-free and ordered delivery** of data.
 - Manages **ports** to direct traffic to the right applications.
 - Uses **TCP (Reliable)** or **UDP (Fast, Unreliable)** for communication.
- **Protocols Used:**
 - **TCP (Transmission Control Protocol)** – Provides reliable, connection-oriented communication.

- **UDP (User Datagram Protocol)** – Provides fast, connectionless communication.
- **TLS (Transport Layer Security)** – Encrypts data for secure transmission.



Layer 4: Application Layer (User Interaction & Data Handling)

- Function:
 - Provides **network services** directly to users and applications.
 - Handles **encryption, authentication, and data encoding**.
 - Supports protocols like HTTP, FTP, SMTP, and DNS.
- Protocols Used:
 - **HTTP/HTTPS** – Web browsing (HTTPS is secure using TLS encryption).
 - **FTP/SFTP** – File transfer (SFTP is secure with SSH).
 - **DNS (Domain Name System)** – Resolves domain names to IP addresses.
 - **SMTP/POP3/IMAP** – Email communication protocols.
- Cybersecurity Concerns:
 - **DNS Spoofing:** Attackers modify DNS records to redirect users to **malicious websites**.
 - **Phishing Attacks:** Fake websites steal user credentials.
 - **MITM Attacks:** Attackers intercept and modify communication between users and websites.
- Security Measures:
 - **DNSSEC (DNS Security Extensions)** – Prevents DNS spoofing.
 - **HTTPS Everywhere** – Enforces encrypted web connections.
 - **Spam Filters & DMARC Authentication** – Protects against email phishing.

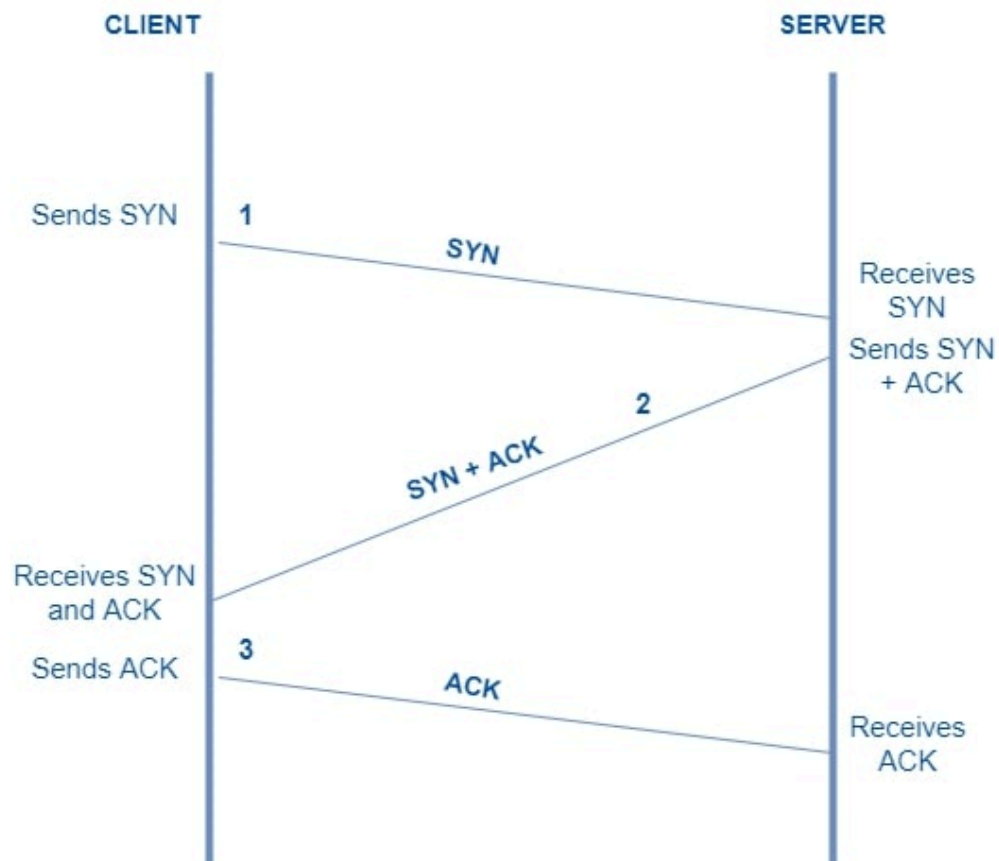
TCP Three-Way Handshake (Connection Establishment)

Before sending data, TCP establishes a **secure connection** between the client and server using the **Three-Way Handshake**

Steps of Three-Way Handshake:

- **1 SYN (Synchronize)** → The client sends a SYN request to the server, asking for a connection.
- **2 SYN-ACK (Synchronize-Acknowledge)** → The server acknowledges with SYN-ACK, confirming it is available.
- **3 ACK (Acknowledge)** → The client sends an ACK, finalizing the connection.

After the handshake, data transfer begins.



- **Cybersecurity Concerns:**
 - **TCP SYN Flood Attack:** Attackers send multiple SYN requests without completing the handshake, leading to DoS attacks.
 - **Session Hijacking:** Attackers steal session information to impersonate users.
 - **Man-in-the-Middle (MITM) Attacks:** Hackers intercept and manipulate TCP sessions.
- **Security Measures:**
 - **SYN Cookies** – Prevents SYN Flood Attacks.
 - **TLS/SSL Encryption** – Secures TCP communication from MITM attacks.
 - **Firewalls & IDS/IPS** – Detect and block malicious TCP packets.

Further Readings:-

- [Phishing Attack](#)
- [TCP/IP](#)
- [Three-way handshake](#)